L1a: Course Orientation, Markets, Assets, and the Decision Problem

CHEME 5660 Cornell University

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we place the course in context, show that you are already a participant in the markets it studies, and close with a question we cannot yet answer rigorously.

Today's concepts

- The structure of the course: how four units connect cash-flow models, uncertainty, contingent claims, and algorithmic decisions.
- Why this concerns you: most U.S. adults hold a retirement account, and the markets those accounts invest in move several hundred times more cash per day than the industries this college trains you to enter.
- The decision problem: an instrument is a schedule of dated payments, and two schedules cannot be ranked without a rule for moving dollars across time.

Note: this lecture has no notebook and no notes packet. These slides are the record.

How This Course Runs

Twenty-eight lectures in four units. Each unit adds one set of tools for the same underlying question, and the six problem sets, spread across the four units, are where you apply them.

What you need

- Materials: each week is posted as a zip file on the repository releases page, or you can clone the repository. Released weeks are frozen: corrections are announced, never made silently.
- Tools: Julia 1.12 through juliaup, Anaconda for Jupyter, and VS Code with the Julia extension. Set these up once, at the start of the semester.
- Questions: ask on the course discussion board rather than by email, so an answer reaches everyone who had the same question.

Policies: grading, attendance, academic integrity, and accommodations are posted in full

What to Expect Each Week

The policies are posted and you can read them. Three things about the weekly work are not in that document.

- Each lecture builds one result and uses it immediately. The material is cumulative, so a missed week makes the next one harder, not just longer.
- You will write code. Not a lot of it, and not from scratch: the course package supplies most of it. But you have to run a model to understand what it does.
- The modeling is harder than the mathematics. Deciding what to model, and admitting when a model is wrong, is what takes the semester.

If you have questions about how the course runs, ask them now.

Most of You Already Hold a Portfolio

Roughly 61% of U.S. adults hold a retirement account: a 401(k), a 403(b), or an IRA. For U.S. households, financial markets are the second largest component of wealth, after real estate. (2025)

If you hold one of those accounts, you are already a market participant.

- That account holds a portfolio: a set of positions, held in some proportion, chosen by someone.
- Choosing those proportions is an allocation decision. It is the decision this course spends four units learning how to make.
- Most account holders have never made that decision explicitly. A default allocation was selected for them and they left it in place.

Leaving the default in place is still an allocation decision.

The Federal Open Market Committee

A committee of twelve people meets eight times a year, and its decisions change the value of everything in that account.

- Thursday: what the committee controls, what it does not, and why that distinction determines which rate you use to value an asset.
- September 15 and 16: the committee meets. We will be three weeks into the course, and we return to the decision in week four with the tools to describe its effect.

Sixty seconds on this today. The details are Thursday's lecture.

Link: FOMC calendar and meeting materials

Market Scale Compared to Two Industries

Two industries this college trains people to enter, compared with the U.S. financial markets. The annual industry figures are divided by 252 trading days.

	Per year	Cash per trading day
Global biopharma	USD 666B	USD 2.64B
Global batteries	USD 181B	USD 0.72B
Together		USD 3.36B
US debt		USD 1,209B
US equity		USD 608B
Options premium		USD 37B
US markets together		USD 1,854B

These markets move about five hundred times more cash per day. The work is engineering: estimation, uncertainty propagation, optimization, and control.

Sources: market sizes 2025 (Fortune Business Insights); turnover 2024–2026 (SIFMA, Cboe)

Premium Paid Versus Value Controlled

The options row is not like the other two. The USD 37B of premium is the cash paid, not the size of the position that cash opens.

Each contract is an option on **100 shares**. At 60.4 million contracts a day, the shares underneath are worth about USD 4 trillion:

$$\underbrace{\text{USD } 37\text{B}}_{\text{premium paid}} \Big/ \underbrace{\text{USD } 4{,}000\text{B}}_{\text{value controlled}} = \underbrace{0.92\%}_{\text{about 92 cents per \$100}}$$

If every contract were exercised on the day it traded, that USD 4 trillion of stock would change hands, which is more than twice the debt and equity rows combined. The previous table reports premium instead, because premium is cash that actually moves.

Careful: $3.9\text{M} \times 100 \times 7{,}692 = \$3.0\text{T}$ of that is one index, settled in cash. It never delivers a share.

The Three Kinds of Asset

Every instrument in this course is one of three kinds. We name them today and examine each one later, at the point in the course where it is needed.

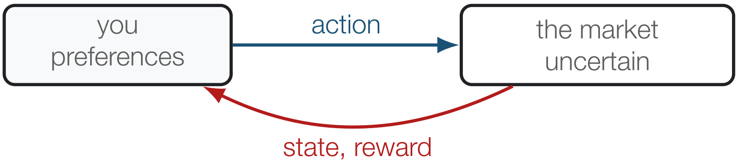
Kind	What you own	Where the course takes it apart
Debt	a promise of dated payments	Unit 1, weeks 2 and 3
Equity	a residual claim on a firm	Unit 2, weeks 3 to 7
Derivative	a claim contingent on another asset	Units 3 and 4, weeks 8 onward

Risk and potential reward both increase down the list. Debt promises a fixed schedule and can fail to pay it. Equity promises nothing and is worth whatever remains after the other claims are paid. A derivative's value depends on the asset underneath it, which gives it the widest range of the three.

Today: we use only the first kind, and we do not use its name until later.

The Decision Problem

All three kinds of asset pose one question. What you observe is the state, what you do is the action, and what comes back is the reward.

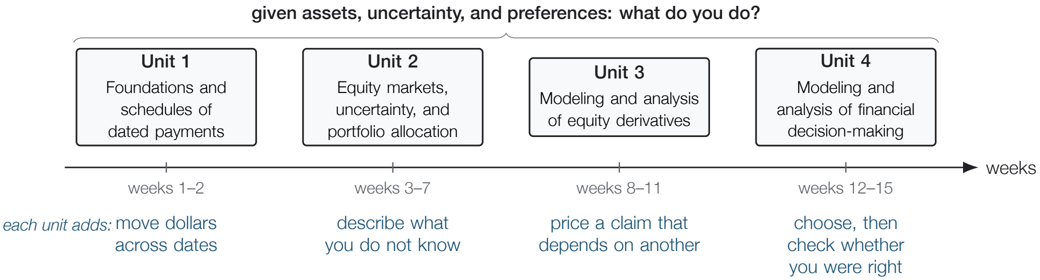


Making this loop operational requires four things: a way to compare payments that arrive on different dates, a way to describe what you do not know, a way to price a claim that depends on another asset, and a rule for choosing. Those are the four units.

Careful: each arrow in the diagram is a modeling choice.

The Four Units of the Course

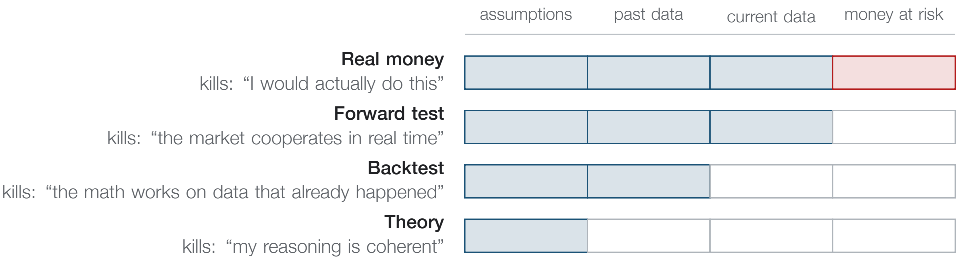
Each unit adds exactly one capability, and the unit that follows it cannot be built without that capability.



Today we start with the leftmost unit, and we stay there through Thursday.

The Validation Ladder

Four ways to check a claim. Each rung uses data the rung below did not see, and each is the only rung that can falsify the claim it tests.



A backtest runs on past data. A forward test runs on actual current data with no money at stake, using a brokerage API such as Alpaca.

Skipping a rung moves the discovery to a stage where money is at stake.

Five Promises for $10,000

Each of these costs $10,000 today, and each is built from a rate the market quoted yesterday. The names come later; for now, only the payments.

- **A** $10,028 back, four weeks from today
- **B** $236 every six months for 10 years, then $10,000
- **C** $264 every six months for **30** years, then $10,000
- **D** $271 every six months for 10 years, then $10,000
- **E** $121 every six months for 10 years, plus an inflation adjustment on every payment and on the $10,000

Which one do you take? Hands up.

As of: published rates for 2026-08-18, each scaled to a $10,000 outlay

What the Five Promises Are Called

	What it is called	Annual rate
A	4-week Treasury bill	3.63%
B	10-year Treasury note	4.71%
C	30-year Treasury bond	5.28%
D	investment-grade corporate bonds	5.42%
E	10-year inflation-indexed Treasury (TIPS)	2.41% real

You compared all five without knowing any of these names, because each name is a label for a schedule of dated payments. Thursday's lecture starts from that definition.

Careful: D is an index of many investment-grade corporate bonds, not one bond. E's rate is *real*, not nominal.

Three Questions We Cannot Answer Yet

Most hands went to **C** or **D**, which is reasonable given the information on the previous slide. That slide left out three things.

- Why not C? It pays the most of the three government rows, but you may need the money before year thirty. If you sell early, you sell at whatever price prevailing rates allow. Week 2.
- Why not D? It pays more than any other row. Those borrowers pay 71 basis points (0.71 percentage points) more than the government at the same ten-year horizon ($5.42\% - 4.71\%$), which is what the market charges for the risk of not being paid and for holding something harder to sell. Units 2 and 3.
- Why not E? It pays the least of any row, but that comparison is not valid: its rate is quoted in different units. Next slide.

Only A, B, and C promise a schedule that is actually certain.

Breakeven Inflation

B and E are the same issuer at the same horizon. B promises nominal dollars; E promises real ones, adjusted for whatever inflation turns out to be. Subtracting one rate from the other leaves the difference between those two units:

$$\underbrace{4.71\%}_{\text{nominal}} - \underbrace{2.41\%}_{\text{real}} = \underbrace{2.30\%}_{\text{breakeven inflation}}$$

That is the ten-year breakeven inflation rate: the inflation rate at which B and E pay you the same amount. The Federal Reserve publishes this rate directly, and its value for the same day is 2.30%. It is not a pure forecast, because it also includes what investors charge for bearing inflation risk.

Market prices contain forecasts, and extracting them is much of what this course does.

Source: FRED series `DGS10`, `DFII10`, `T10YIE`, as of 2026-08-18

A Bank Failure in March 2023

In March 2023 a bank with about $200 billion in assets failed in two days. It held one of the five promises on your list.

Which one, and why?

None of the five defaulted. Every payment that came due was made as promised, and the bank failed anyway.

The clip identifies which one. It does not explain why in a way you can check; week 2 supplies that explanation.

Clip: PBS NewsHour, 28 March 2023, 6 minutes

Why You Cannot Rank the Five Yet

Every promise on that list is issued by the United States government or by large companies that have paid their obligations for decades. No vote in this room was careless.

You cannot rank those five yet, and the reason is precise:

> You have no rule for comparing dollars that arrive on different dates.

$264 in thirty years and $10,028 in four weeks cannot be compared until we state what a dollar on one date is worth on another. We build that rule on Thursday, and it is the first thing this course does.

Summary

This lecture placed the course in context, showed that you are already a market participant, and read five financial promises as schedules of dated payments.

Key takeaways

- **You are already a portfolio holder.** Retirement accounts make market participation the default, so nearly everyone answers the allocation question, most of them by leaving a default in place.
- **An instrument is a schedule.** A bill, a note, a bond, a corporate bond, and an inflation-indexed security are labels for schedules of dated payments, though not every schedule is equally certain.
- **Validation happens in stages.** Theory, backtest, forward test, and real money each use different data, and each is the only stage that can falsify the claim it tests.

On Thursday we build the rule that lets us rank those five schedules.

Further Reading

Not assigned, and not on any problem set. These four sources describe the Federal Reserve System, the committee mentioned earlier, and the Treasury.

- *The Structure and Functions of the Federal Reserve System*, Federal Reserve Board: the roles of the Board of Governors, the Reserve Banks, and the FOMC.
- *Economy at a Glance: Policy Rate*, Federal Reserve Board: the target range, the effective rate, and the current implementation tools.
- *Fedpoints: Federal Reserve System*, Federal Reserve Bank of New York: an institutional overview.
- *Role of the Treasury*, U.S. Department of the Treasury: fiscal, financial, and debt-management responsibilities. The Treasury is a different institution from the Federal Reserve; week 2 distinguishes them.